Re-analysis Summary Report for Peer-evaluation

Paper ID: **Hurst_EvoHumanBehavior_2017_yypJ**

Re-analyst's ID: **ccbe4**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
… those with a faster life strategy report greater levels … of psychopathology (p. 1.)

The corresponding article

Here you can find the original article: https://osf.io/tqmbe/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/vhmgc/?view_only=5c6bedb36b2549a88ac137d6c746bcb8

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
To begin, I reproduced the partial correlation analyses from the original manuscript. To do so, I specified an observed-variable path model in lavaan (Rosseel, 2012), where controlled for age in the variables of interest. I then combined these models into a single path model where I included the covariance between the two measures of life-history. I found support for the claim "those with a faster life strategy report greater levels … of psychopathology" consistent with the original paper, where the Mini-K (Figueredo et al., 2006; $r = -0.409$, $p < 0.001$) and High-K Strategy Scale (HKSS; Giosan, 2006; $r = -0.513$, $p < 0.001$) measures showed significant negative partial correlations with total psychopathology symptoms (DSM-5 Self-Rated Level 1 Cross-Cutting Symptom Measure-Adult; American Psychiatric Association, 2013), controlling for age. The two life history measures showed a significant positive partial correlation ($r = 0.669$, $p < 0.001$), again controlling for age.

I then performed two sensitivity analyses to assess the robustness of these initial analyses. The first addressed the use of the two individual measures of life history. I extracted the first principal component from the two measures to ensure they were assessing similar variance in a follow-up partial correlation analysis. Not only did the measures correlate strongly with the PCA component (standardized loading = 0.92), but the composite measure also showed a partial relationship with total psychopathology symptoms similar to the individual measures (r = -0.504, $p < 0.001$). The second sensitivity analyses addressed the limited control for additional variable relationships in the partial correlation analysis. Rather than only controlling for age, I constructed a partial correlation network, where I controlled for the remaining variables (11 in total; see supplemental materials) in the dataset and re-assessed the primary claim. Controlling for multiple comparisons (Bonferoni correction: 0.05 / 55 = 0.0009), total psychopathology symptoms retained a significant negative partial correlation with the Mini-K (r = -0.303, p = 0.0005) but not the HKSS (r = 0.010, p = 0.912) when controlling for all other variables. The positive correlation between life-history measures also remained significant (r = 0.556, $p < 0.0001$). These sensitivity analyses lend additional support to the central claim in the manuscript.

Rosseel Y (2012). "lavaan: An R Package for Structural Equation Modeling." Journal of Statistical Software, 48(2), 1–36. doi: 10.18637/jss.v048.i02.

Figueredo, AJ, Vásquez, G, Brumbach, BH, & Schneider, SMR (2004). The heritability of life history strategy: The k-factor, covitality, and personality. Biodemography and Social Biology,51,121–143.http://dx.doi.org/10.1080/19485565.2004.9989090.

Giosan, C (2006). High-K strategy scale: A measure of the high-K independent criterion of fitness. Evolutionary Psychology, 4, 394–405.

American Psychiatric Association (2013b).The DSM-5 self-rated level 1 cross-cutting symptom measure - adult.

The analyst's conclusion: In both the main analyses and follow-up sensitivity analyses, I found support for the claim "those with a faster life strategy report greater levels … of psychopathology".

The analyst's categorization of the result:
The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should report the Mini-K measure instead of HKSS for life history strategy.

Here, the analyst reported the following steps and results of the analysis:
I tested the partial correlation of the life history strategy (indexed by the Mini-K; Figueredo et al., 2006) with total psychopathology (DSM-5 Self-Rated Level 1 Cross-Cutting Symptom Measure-Adult; American Psychiatric Association, 2013), controlling for age. To do so, I specified an observed-variable path model in lavaan (Rosseel, 2012), where I controlled for age in the analyses of interest. I found support for the claim "those with a faster life strategy report greater levels … of psychopathology" (rpartial = -0.409, $p < 0.001$, N = 138) consistent with the original paper.

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/ktr9u/